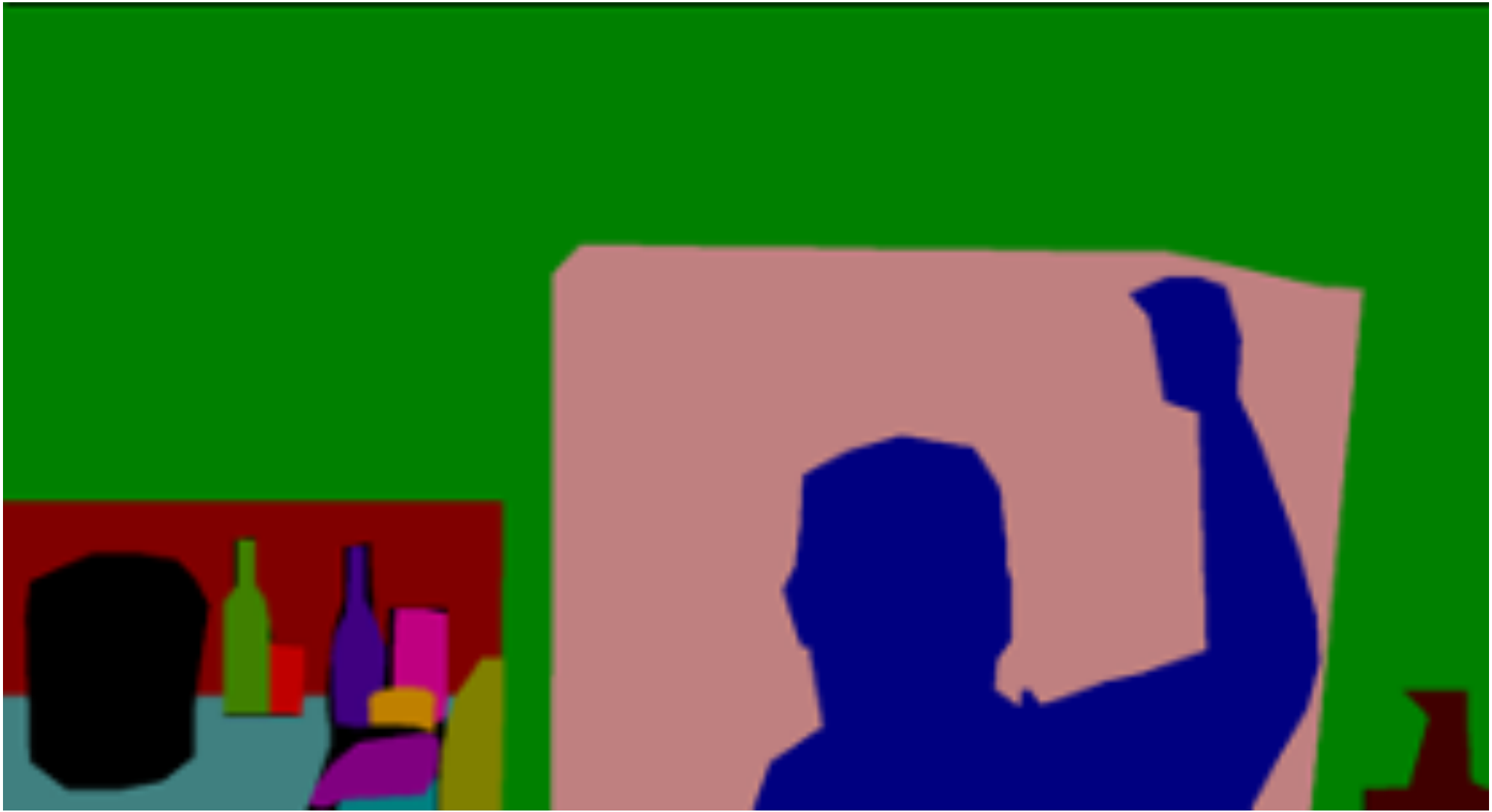
User #3:

Segment all objects



User #4:
Segment the object that aligned with the text prompt

Text prompt: A frog is holded by a person in his hand and ....

User #1:

Segment all objects

UniVS: Video Segmentation Foundation Model

Minghan Li. **UniVS:** Unified Video Segmentation with Prompts (CVPR 2024)

Input

Output

User #2:

Segment the cat

Category names:

Person, Roof, Dog, Washer, ...

100%

DIGITAL

UniVS: Video Segmentation Foundation Model

User #1:
Segment all objects

User #4:
Segment the object that aligned with the text prompt

User #3:
Segment all objects

User #2:
Segment the cat

Input

Category names:
Person, Roof, Dog, Washer, …

Text prompt: A frog is holded by a person in his hand and ….

UniVS
Video Segmentation Foundation Model

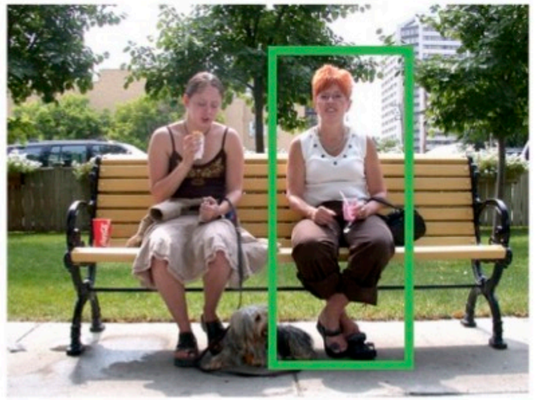
Output

Minghan Li. **UniVS:** Unified Video Segmentation with Prompts (CVPR 2024)

Two Key Challenges in Foundation Model

Conflicts across Different Tasks

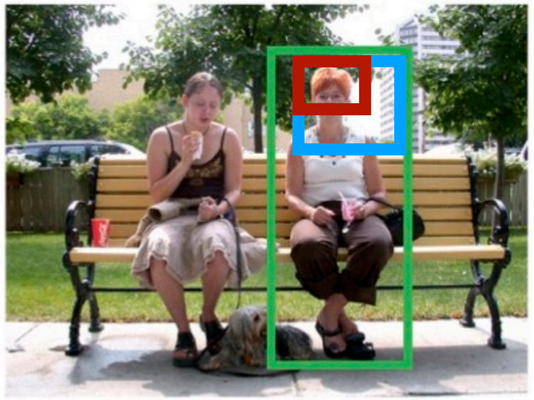
Semantic Consistency



Category name: Woman

Description:
Woman with white T-shirt

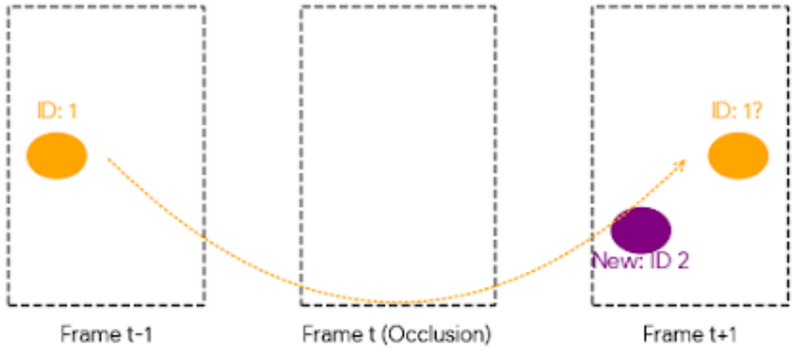
Object Granularity



Person vs. Face vs. Hair

Complex Tracking Trajectory

Temporal Consistency



Re-appear or newly-appear objects

Minghan Li. **UniVS:** Unified Video Segmentation with Prompts (CVPR 2024)